

Survival analysis within infectious disease modelling

From local association in bivariate event time data to the analysis of multivariate serological data

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Infectious disease epidemiology

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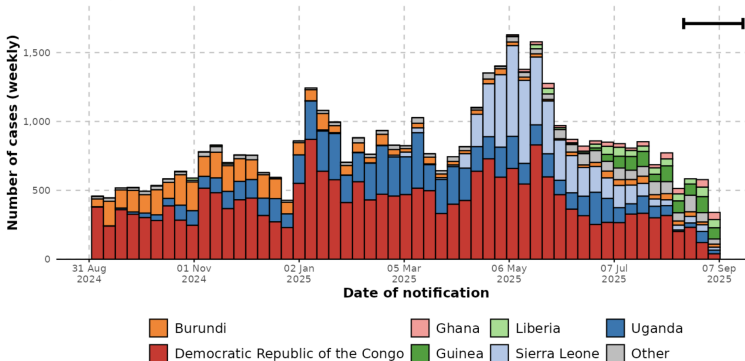
News | Health

Ebola outbreak: the data that show why researchers are so alarmed

The size of the outbreak in its initial days is worrying researchers. The next few weeks will determine how large it grows, they say.

Trends in confirmed mpox cases

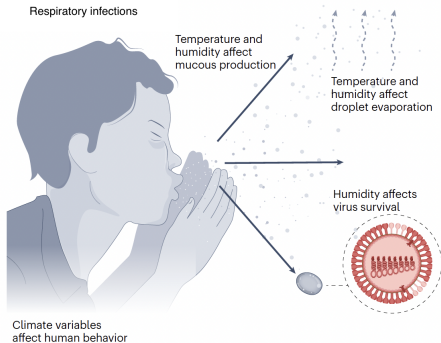
Bracket at end of curve indicates potential reporting delays in recent weeks of data.
Data as of 07 Sep 2025



WHO says 12th person infected with hantavirus detected in the Netherlands

WHO chief, Tedros Adhanom Ghebreyesus, urged all countries to monitor passengers who were on the MV Hondius cruise ship.

How are infectious diseases transmitted?



Vectorborne diseases

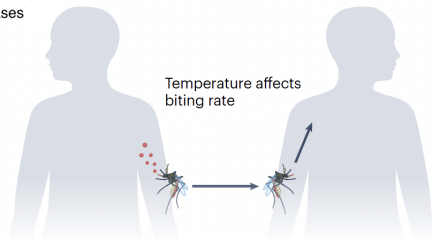


Figure: Respiratory disease transmission through direct or indirect droplet contact (left panel). Vectorborne disease spread through, for example, mosquitos.
Source: *Adapted from Baker et al. (2026).*

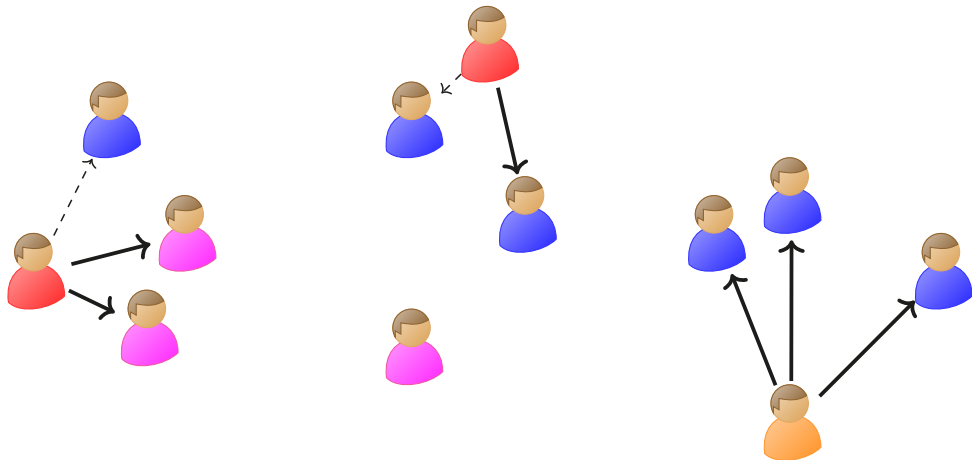
How is disease transmission described?

- ▶ **Infectious disease epidemiology** relies on statistical and mathematical approaches to model the spread of infectious diseases (Anderson and May, 1991)
- ▶ **Transmission process** governed by **(social) mixing patterns** (with contacts deemed relevant for transmission)
- ▶ **Transmission parameters** estimated from **infectious disease data** (serological data, outbreak data, etc.), e.g. force of infection λ , basic reproduction number R_0 , ...
- ▶ **Force of infection** relates to transmission rates through the so-called “**mass action principle**”:

$$\lambda = \beta I$$

Heterogeneity in the acquisition of infections

- Individual variation in **susceptibility to infection**, **infectiousness upon infection** (e.g., superspreaders and viral load) and **social contact behaviour**



- ▶ Variation in **measurable** (e.g., age) and **unmeasurable** (unobserved) (resulting from, e.g., genetic predisposition) factors
- ▶ Impact on **key epidemiological parameters** studied before by, e.g., Coutinho et al. (1999) and Farrington et al. (2001)
- ▶ Recently, investigation of **types of superspreading** on SARS-CoV-2 infections (e.g., Kuylen et al., 2022)
- ▶ Sources of heterogeneity are **not always accounted for** in (statistical or mathematical) infectious disease models
- ▶ **Goal:** Use of **survival models** to accommodate (individual) heterogeneity in infectious disease data and to study **routes of transmission** for paired (serological) data

Frailty models to reveal how infectious diseases are transmitted

Assume that for each individual we have

- ▶ **Measured characteristics** $\mathbf{x} = (x_1, \dots, x_p)^T \in \mathbb{R}^{p+1}$
 - ▶ age
 - ▶ sexual behaviour
 - ▶ needle sharing practice
- ▶ **Unobserved (latent) characteristics** $\mathbf{z} = (z_1, \dots, z_q)^T \in \mathbb{R}^{q+1}$
 - ▶ susceptibility
 - ▶ infectiousness
 - ▶ sociability and social environment
 - ▶ behavioural variables (e.g., propensity to use public transport or attend large gatherings)

- ▶ Extended effective contact rates $\beta(\mathbf{x}, \mathbf{z}; \mathbf{x}', \mathbf{z}')$ represent the **per capita rate** at which an infectious individual with characteristics $(\mathbf{x}, \mathbf{z})$ **makes effective contacts** with susceptible individuals with characteristics $(\mathbf{x}', \mathbf{z}')$ (Diekmann and Heesterbeek, 2001)
- ▶ **Multiplicative decomposition** (Farrington et al., 2001) and **separability**:

$$\beta(\mathbf{x}, \mathbf{z}; \mathbf{x}', \mathbf{z}') = \alpha(\mathbf{x}, \mathbf{z}; \mathbf{x}', \mathbf{z}') \beta_0(\mathbf{x}, \mathbf{x}') = w(\mathbf{x}, \mathbf{z}) \beta_0(\mathbf{x}, \mathbf{x}') w'(\mathbf{x}', \mathbf{z}'),$$

where

$$\beta_0(\mathbf{x}, \mathbf{x}') = \int_{\mathbf{z}} \int_{\mathbf{z}'} \beta(\mathbf{x}, \mathbf{z}; \mathbf{x}', \mathbf{z}') f_{\mathbf{Z}}(\mathbf{z}) f_{\mathbf{Z}'}(\mathbf{z}') d\mathbf{z} d\mathbf{z}'$$

is the **average effective contact rate** for individuals with measured characteristics $\mathbf{x}$ and $\mathbf{x}'$

- ▶ Functions $w(\cdot, \cdot)$ and $w'(\cdot, \cdot)$ might differ and, for reasons of identifiability, we assume

$$E_{\mathbf{z}} [w(\mathbf{x}, \mathbf{z})] = E_{\mathbf{z}'} [w'(\mathbf{x}, \mathbf{z}')] = \tilde{w}(\mathbf{x})$$

- For infections in endemic equilibrium, **force of infection** can be expressed as **integral equation**:

$$\begin{aligned}\lambda(\mathbf{x}, z) &= \int_{\mathbf{x}'} \int_{z'} \beta(\mathbf{x}, z; \mathbf{x}', z') I(\mathbf{x}', z') dz' d\mathbf{x}' \\ &= w(\mathbf{x}, z) \int_{\mathbf{x}'} \int_{z'} \beta_0(\mathbf{x}, \mathbf{x}') w'(\mathbf{x}', z') I(\mathbf{x}', z') dz' d\mathbf{x}' =: w(\mathbf{x}, z) \lambda_0(\mathbf{x}),\end{aligned}$$

with $\lambda_0(\mathbf{x})$ the so-called **baseline force of infection**

- This defines a **frailty model** for the equilibrium force of infection
- Different **sources of unobserved heterogeneity**, expressed in terms of $w(\mathbf{x}, z)$ and $w'(\mathbf{x}', z')$, and their impact on, e.g., R_0 can be studied
- **Proportional hazards model**: $w(\mathbf{x}, z) = z\tilde{w}(\mathbf{x}) = z \exp(\beta^T \mathbf{x})$

- ▶ Bivariate serological data $(y_1, y_2, \mathbf{x})$ under endemic equilibrium with y_i ($i = 1, 2$) representing the immunological status at sampling time
- ▶ Solving the system of (ordinary) differential equations (under endemic equilibrium) provides a (numerical) solution for $I(\mathbf{x}', \mathbf{z}')$
- ▶ Marginalisation of $I(\mathbf{x}', \mathbf{z}')$ (or equivalently $S(\mathbf{x}', \mathbf{z}')$) in terms of frailty distribution (characterised by $f_{\mathbf{Z}}(\mathbf{z})$) leads to (tractable) expressions used in ML inference
- ▶ Extension to bivariate setting with infection-specific hazard functions ($i = 1, 2$):

$$\lambda_i(\mathbf{x}, \mathbf{z}_i) = w_i(\mathbf{x}, \mathbf{z}_i) \lambda_{i0}(\mathbf{x})$$

- ▶ Different types of frailty models induced by choice of bivariate frailty distribution $f_{\mathbf{Z}_1, \mathbf{Z}_2}$
 - ▶ Shared frailty model structure: $\mathbf{Z}_1 = \mathbf{Z}_2 = \mathbf{Z}$ (heterogeneity in contact rates)
 - ▶ Correlated frailty model structure: $\mathbf{Z}_1, \mathbf{Z}_2$ (heterogeneity in susceptibility & infectiousness)

Measuring local association between infection times

- ▶ In infectious disease epidemiology, when unraveling transmission routes, one is often interested in studying **association between infection times T_1 and T_2**
- ▶ Strength of association can be expressed in terms of **global and local association measures**
- ▶ **Cross-ratio function** was introduced by Clayton (1978):

$$\begin{aligned}\theta(t_1, t_2) &= \frac{\lambda_1(t_1|T_2 = t_2)}{\lambda_1(t_1|T_2 > t_2)} = \frac{\lambda_2(t_2|T_1 = t_1)}{\lambda_2(t_2|T_1 > t_1)} \\ &= \frac{S(t_1, t_2)\partial_{12}^2 S(t_1, t_2)}{\partial_1^2 S(t_1, t_2)\partial_2^2 S(t_1, t_2)}\end{aligned}$$

- ▶ **Interpretation** (Oakes, 1982, 1986):
 - ▶ $\theta(t_1, t_2) = 1$: infection times T_1 and T_2 are **independent**
 - ▶ $\theta(t_1, t_2) > 1$: infection times T_1 and T_2 are **positively associated**
 - ▶ $\theta(t_1, t_2) < 1$: infection times T_1 and T_2 are **negatively associated**

- **Conditional frailty distribution**, i.e., the distribution of Z conditional on no infection at least up to time t , is characterised by the density function

$$f_Z(z | T > t) = \frac{S(t | z) f_Z(z)}{S(t)},$$

where f_Z is the (unconditional) probability density function of individual frailty Z

- **Relative frailty level**, defined as

$$Z_t = \frac{Z}{E(Z | T > t)},$$

gives rise to the so-called **relative frailty distribution** in survivors up to time t

- ▶ Relative frailty variance is equal to

$$\text{RFV}(t) = \text{Var}(Z_t \mid T > t) = \frac{\text{Var}(Z \mid T > t)}{E(Z \mid T > t)^2}$$

- ▶ Different choices of the frailty distribution f_Z lead to different expressions of the $\text{RFV}(t)$
- ▶ Extension for two infection times T_1 and T_2 and shared frailty level Z :

$$\text{RFV}(t_1, t_2) = \frac{\text{Var}(Z \mid T_1 > t_1, T_2 > t_2)}{E(Z \mid T_1 > t_1, T_2 > t_2)^2}$$

- ▶ One can show that (Farrington et al., 2012):

$$\theta(t_1, t_2) = \text{RFV}(t_1, t_2) + 1$$

- ▶ Generalisation for correlated frailty models possible in terms of $\text{RFCV}(t_1, t_2)$

- ▶ Cross-ratio function can be considered for **diagnostics regarding frailty distribution** (or copula function) used to describe dependence in bivariate survival data
- ▶ **Some earlier work:**
 - ▶ Hsu and Prentice (1996): crude nonparametric estimator for the cross ratio function
 - ▶ Nan et al. (2006): parametric piecewise constant estimation in rectangular regions
 - ▶ Li and Lin (2006) & Li et al. (2008): copula-based approaches
 - ▶ Glidden (2007): implicit nonparametric estimator for the cross ratio function, though very rough behaviour
 - ▶ Hu et al. (2011): parametric polynomial models for the (log-transformed) cross ratio function applied to right-censored bivariate survival data
- ▶ **Our work:**
 - ▶ Smooth Bernstein-based estimator of conditional hazards in $\theta(t_1, t_2)$ (Abrams et al., 2020)
 - ▶ Direct Bernstein-based estimation of survival function and its derivatives in $\theta(t_1, t_2)$ (Abrams et al., 2024)

- ▶ Bernstein-based estimators for $\theta(t_1, t_2)$ under different types of (right-)censoring
- ▶ **Semi-parametric estimators based on P- and O-splines** accounting for boundary effects (**joint work with Miguel Beynaerts, Niel Hens and Anneleen Verhasselt**)
- ▶ **Goodness-of-fit measures** based on $\theta(t_1, t_2)$ to assess the choice of (bivariate) frailty distribution
- ▶ Serological data: estimation of $\theta(t_1, t_2)$ under Type I interval-censored (current status) data and inferring time of infection from serological data

- ▶ Consider a sample $(X_{1i}, \Delta_{1i}, X_{2i}, \Delta_{2i})$, $i = 1, \dots, n$, of bivariate time-to-event data from $(X_1, \Delta_1, X_2, \Delta_2)$ with for $j = 1, 2$:
 - ▶ $X_j = \min(T_j, C_j)$ **observation times**,
 - ▶ $\Delta_j = I(T_j \leq C_j)$ **censoring indicators**, and
 - ▶ (C_1, C_2) **bivariate censoring times**,under **independent (and non-informative) right censoring**
- ▶ Hu et al. (2011) proposed a **pseudo-partial likelihood** approach for the estimation of the CRF by considering T_1 as a covariate in a Cox PH model
- ▶ **Ratio of hazards** of event two in groups $\{i \mid T_{1i} = t_1\}$ and $\{i \mid T_{1i} > t_1\}$ equals the CRF

- ▶ CRF is **symmetric** with respect to switching the roles of T_1 and T_2 : symmetrization step in **log-likelihood** to guarantee symmetry for the estimator:

$$l(\beta) = \sum_i \left[l_i^{(1)}(\beta) + l_i^{(2)}(\beta) \right], \quad (1)$$

where β are parameters associated with the $\log \theta(t_1, t_2)$

- ▶ Pseudo-partial log-likelihood function (1) is maximized to obtain the Maximum Likelihood (ML) estimates $\hat{\beta} = \operatorname{argmax}_{\beta} l(\beta)$
- ▶ **Tensor product spline approach:**

$$\log \theta(t_1, t_2) = \sum_{r=1}^{r'} \sum_{s=1}^{s'} \beta_{rs} B_1^r(t_1) B_2^s(t_2)$$

Flexible estimation of $\log \theta(t_1, t_2)$ requires a large parameter space. For parameter identifiability, we introduce **penalization** into the (log-)likelihood

$$l_{\lambda}(\beta) = l(\beta) - \frac{\lambda_1}{2} \beta^T S_1 \beta - \frac{\lambda_2}{2} \beta^T S_2 \beta$$

with

- ▶ $\lambda = (\lambda_1, \lambda_2)$ a vector of smoothing parameters to be chosen in a data-driven way, and
- ▶ S_i ($i = 1, 2$) the corresponding positive semidefinite penalty matrices

Possible penalties:

- ▶ **Discrete (d -order difference) penalties:** P-splines (Eilers & Marx, 1996)
- ▶ **Integrated squared derivative penalties:** O-splines (O'Sullivan, 1986)

General idea from Wahba (1983):

$$\beta|\lambda \sim N\left(\mathbf{0}, \left(\sum_k \lambda_k \mathbf{S}_k\right)^{-1}\right).$$

This leads to the Laplace approximation of the (penalized) log-likelihood

$$\mathcal{V}(\lambda) = l(\beta) - \frac{1}{2}\beta^T S_\lambda \beta + \frac{1}{2} \log |S_\lambda|^+ - \frac{1}{2} \log |\mathcal{H}_\lambda| + \frac{\mathcal{M}_p}{2} \log(2\pi),$$

where $S_\lambda = \sum_i \lambda_i \mathbf{S}_i$, $|\cdot|^+$ the product of all positive eigenvalues, $\mathcal{H}_\lambda$ the negative Hessian of the penalized log-likelihood and $\mathcal{M}_p$ the number of zero eigenvalues of S_λ when all λ_i are strictly positive

Wood and Fasiolo (2017) proposed a **simple iterative procedure** to maximize the **Laplace approximated (penalized pseudo-partial) log-likelihood function**

$$\lambda_i^* = \frac{\text{tr}(S_{\lambda}^{-1} S_i) - \text{tr}(\mathcal{H}_{\lambda}^{-1} S_i)}{\hat{\beta}^T S_{\lambda} \hat{\beta}} \lambda_i.$$

Procedure:

- 1 Choose initial $\lambda_{(0)}$
- 2 Maximize penalized likelihood to obtain $\hat{\beta}|\lambda_{(0)}$
- 3 Update $\lambda_{(0)}$ to $\lambda_{(1)}$ using $\hat{\beta}|\lambda_{(0)}$
- 4 Re-estimate $\hat{\beta}|\lambda_{(1)}$
- 5 Check convergence criteria
- 6 Repeat steps 2–5 until convergence

We fit P-splines with dimension 17×17 to simulation sets of size 500

Simulation setting:

- ▶ Dependence through **Frank copula** with association parameter $\alpha = 3$: $(u_1, u_2) \sim \mathcal{C}_\alpha$
- ▶ Implied **Kendall's tau** $\tau_K \approx 0.3$
- ▶ **Uniform margins** $F_T \sim \text{Unif}(0, 5)$ such that $(T_1, T_2) = (F_T^{-1}(u_1), F_T^{-1}(u_2))$
- ▶ **Independent bivariate censoring times** $C_j \sim \text{Exp}(\lambda_C)$ ($j = 1, 2$), with λ_C chosen to have 35% or 60% of censoring
- ▶ $N = 500$ **simulation runs**

Table: P-spline estimates of $\log \theta$ under Frank dependence with $\tau = 0.31$ for 35% and 60% censoring.

T_1/T_2		50%		
	Censoring	n	$\log \theta$	$\log \hat{\theta}$ (SE_e , SE_m , CP)
50%	35%	500	0.462	0.466 (0.188, 0.182, 93.8%)
		750		0.457 (0.149, 0.144, 95.4%)
		1000		0.456 (0.116, 0.124, 95.2%)
	60%	500		0.467 (0.255, 0.254, 92.8%)
		750		0.455 (0.193, 0.199, 93.4%)
		1000		0.458 (0.175, 0.165, 94.6%)

- Estimator performs well in terms of bias and CP in regions with sufficient uncensored observations (example at the main diagonal)
- Few uncensored observations in boundary area reduce finite-sample performance (compared to more parsimonious parametric alternatives)

Selected results: Frank copula with P-splines

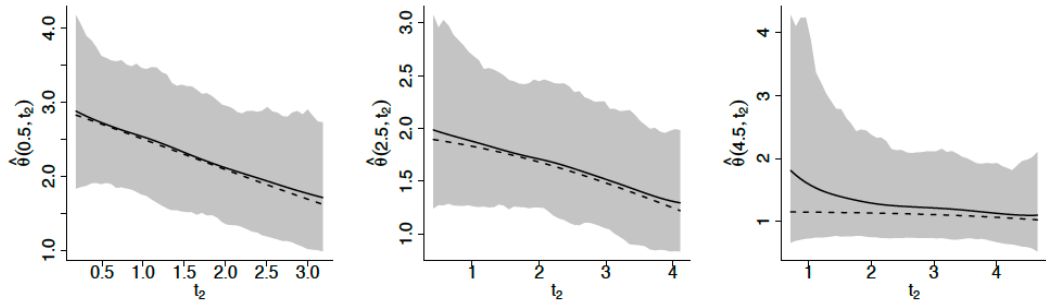


Figure: P-spline fit with 35% censoring for $n = 500$.

Back-calculating bivariate infection times

- ▶ In **cross-sectional serosurvey data**, often univariate monitoring times are present (e.g., multiplex or ELISA testing based on blood sample collected at single time point)
- ▶ Consequently, only information available with regard to $S(t, t)$
- ▶ **Problem:** CRF cannot be directly estimated from bivariate serological (current status) data in case of univariate monitoring times
- ▶ **Solution:** Combine (limited) **longitudinal antibody kinetics data** with cross-sectional serological survey data

- **Longitudinal antibody model:** Let t_{ij}^* refer to the time since infection at measurement occasion $j = 1, \dots, J$, for an infected and seropositive individual $i = 1, \dots, n_s$. The continuous (log-transformed) antibody titer Y_{ijk}^* with respect to pathogen $k = 1, 2$ for the j th measurement of individual i is (assumed to be) a realization from

$$Y_{ijk}^* \sim N(f_k(t_{ij}^* | \tilde{\Theta}_k), \sigma_k^2),$$

with $f_k(t_{ij}^* | \tilde{\Theta}_k)$ describing the antibody kinetics curve for pathogen k

- **Antibody kinetics curve:** several (parametric) choices can be considered
- **Beta distribution** for increasing and falling antibody levels with time since infection
 - **Exponential distribution** reflecting exponential decay with time since infection

$$f_k(t_{ij}^* | \tilde{\Theta}_k) = b_k a_k \exp(-a_k t_{ij}^*)$$

- ▶ **Cross-sectional serological data:** Assuming the presence of lifelong humoral immunity after infection, and pathogen dynamics being in endemic equilibrium, with force of infection

$$\lambda_k(a) = (\alpha_k a - \gamma_k) e^{-\beta_k a} + \gamma_k,$$

- ▶ The seroprevalence for pathogen k is then given by

$$\pi_k(a) = 1 - \exp \left\{ \frac{\alpha_k}{\beta_k} a e^{-\beta_k a} - \left(\frac{\alpha_k}{\beta_k} - \gamma_k \right) \frac{1}{\beta_k} (e^{-\beta_k a} - 1) - \gamma_k a \right\}.$$

Joint population and antibody kinetics (Bayesian) model

- ▶ Joint estimation of
 - ▶ the population model parameters $\Phi = (\Phi_1, \Phi_2)$ with $\Phi_k = (\alpha_k, \beta_k, \gamma_k)$,
 - ▶ the antibody kinetics model parameters $\Theta = (\Theta_1, \Theta_2)$ with $\Theta_k = (a_k, b_k, \sigma_k^2)$, and
 - ▶ the infection times $T = (T_1, T_2)$ with $T_k = (T_{1k}, \dots, T_{K,k})$
- ▶ The following **posterior density** for (T, Θ, Φ) is then considered

$$f(T, \Theta, \Phi \mid \mathbf{y}^*, \mathbf{t}^*, \mathbf{y}, \mathbf{z}, \mathbf{a}) \propto \underbrace{\left(\prod_{i=1}^n f(\mathbf{y}_{i.1}^*, \mathbf{y}_{i.2}^* \mid \mathbf{t}_{i.1}^*, \mathbf{t}_{i.2}^*, \Theta) \right)}_{\substack{\text{Antibody kinetics model} \\ \text{(longitudinal samples)}}} \underbrace{\left(\prod_{l=1}^K g(\mathbf{y}_l, a_l, \mathbf{Z}_l \mid T_l, \Theta, \Phi) \right)}_{\text{Population model}} \\ \times \underbrace{P(T)P(\Theta)P(\Phi)}_{\text{Priors}},$$

where $g(\mathbf{y}_l, a_l, \mathbf{Z}_l \mid T_l, \Theta, \Phi)$ represents **likelihood contributions** based on the serostatus Z_1 and Z_2 , accommodating potential **right censoring in infection times**

- ▶ **Inferred times since infection** for individual l and pathogen k is then $\tau_{lk} = a_l - T_{lk}$.

- ▶ Longitudinal antibody data:
 - ▶ Exponential decay of antibody levels following infections
 - ▶ Five longitudinal antibody titer measurements ($n_s = 5$) for five individuals
- ▶ Cross-sectional serological data:
 - ▶ Bi-exponential force of infection for two pathogens
 - ▶ Uniform sampling times in cross-sectional sample
 - ▶ Independent infection times related to the two pathogens
 - ▶ Sample size of $K = 500$ individuals in the generated cross-sectional serological data

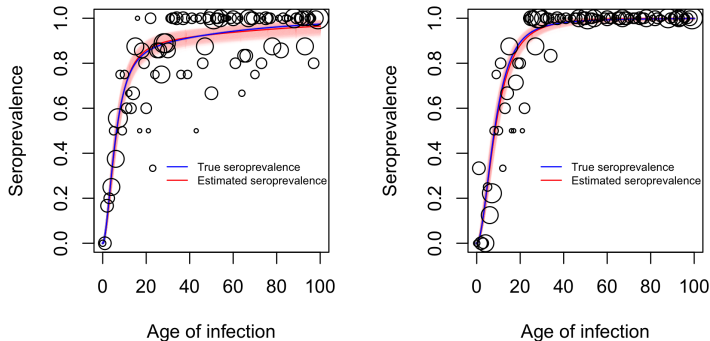


Figure: Posterior median of the seroprevalence for pathogen 1 (left panel) and pathogen 2 (right panel) (red solid lines) with 95% credible intervals (pink shaded areas). True seroprevalence in blue.

Inferred infection times (age of infection)

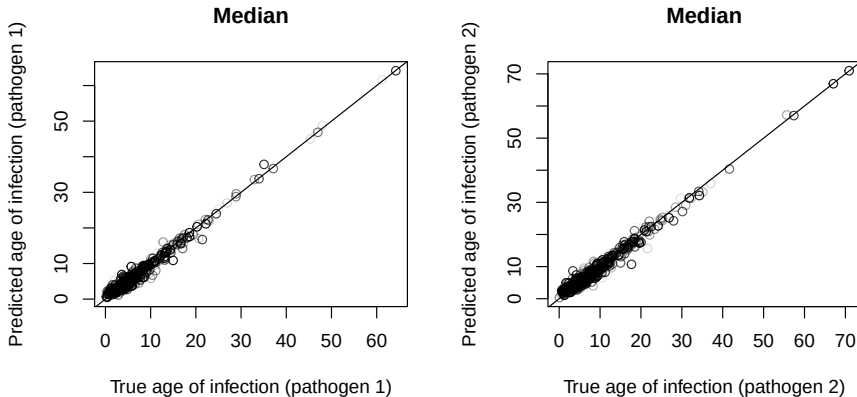


Figure: Posterior median of the ages at infection with pathogen 1 (left panel) and pathogen 2 (right panel) plotted versus the true ages at infection.

CRF estimation using P-spline model

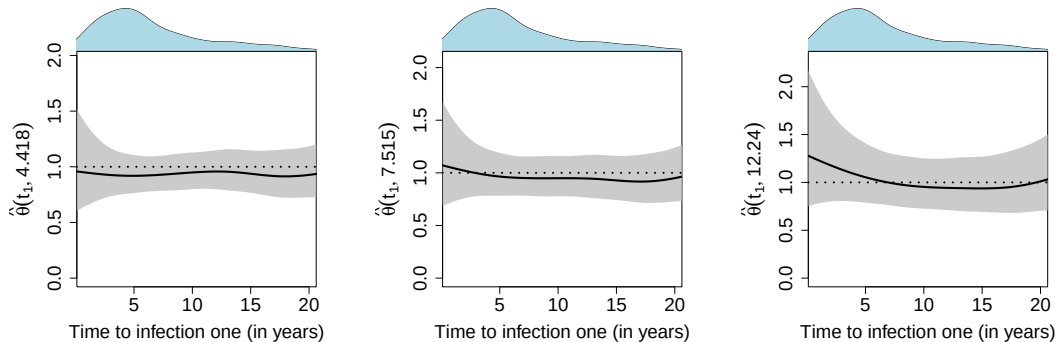


Figure: P-spline fit to the inferred infection times based on the Bayesian model for combined antibody titer data and cross-sectional population-level serology.

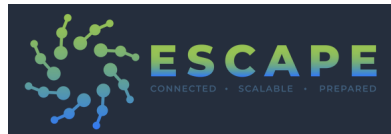
Conclusions

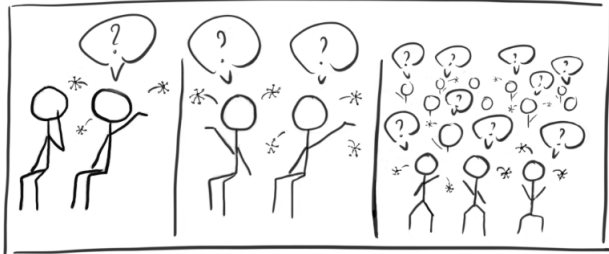
- ▶ More complex longitudinal profiles for **antibody kinetics** with association across repeated measurements and between pathogen-specific antibody titers
- ▶ Flexible modeling of the underlying force of infection for pathogens (instead of bi-exponential function)
- ▶ Different association structures imposed through **conditional frailty approach** (implying mechanistic MAP interpretation)
- ▶ **Real-life data analysis**: dengue serotype data from Cambodia, Southeast Asia

- ▶ **Serial serological survey data** collected at different cross-sectional time points: estimation of age-time specific force of infection (Nagelkerke, 1999)
- ▶ Extension to encompass **(time-varying) individual heterogeneity** in acquisition of infections
- ▶ **Sero-surveillance** data used to assess and **quantify outbreak risk** (Abrams et al., 2014, Hens et al., 2015): new approaches for vaccine-preventable diseases (e.g., measles).

Acknowledgements

All collaborators, SIMID members and co-authors





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